

PROOF OF CONCEPT

Vitsika

Malagasy ant identification from a single photograph

vitsika is Malagasy for ant. Genus-level, 27 genera, built on public data and a frozen public model.

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Vitsika Malagasy ant genus identification · proof of concept

[← Identify another specimen](#)

QUERY IMAGE



casent0002219_p.jpg · embedded in 1102 ms

GENUS PREDICTION

Royidris 37%
subfamily myrmicinae

✓ Held-out test specimen casent0002219

① Classifier confidence is modest (37%) for *Royidris* ([see below](#)).

1. **Royidris** myrmicinae
2. **Nesomyrmex** myrmicinae
3. **Pheidole** myrmicinae

[See where this lands on the atlas →](#)

[← Identify another specimen](#)

hf-hub:imageomics/bloclip-2 · probe 2026-08-29
genera, not a calibrated chance of being right.

Most similar training specimens

cosine similarity of BloClip



Royidris notorthotenes
myrmicinae



Royidris robertsoni
myrmicinae



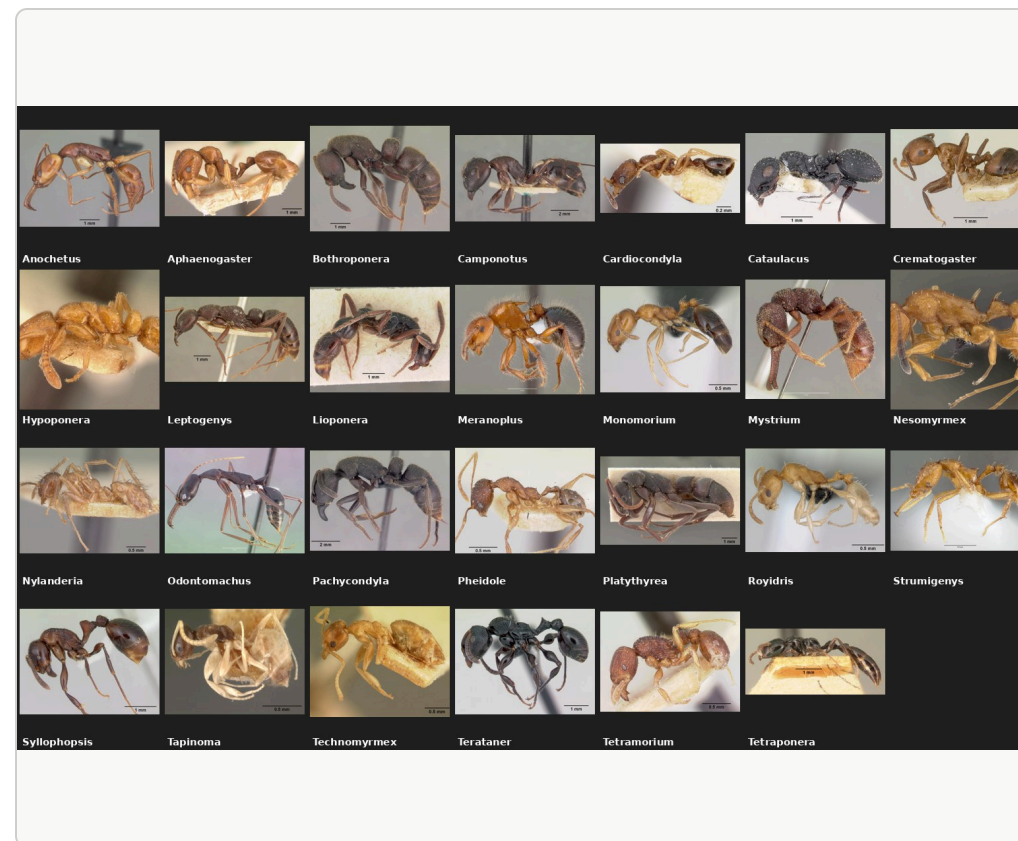
Royidris notorthotenes
myrmicinae



Monon bificidoc

Identification is the bottleneck, and it happens far from the field

- AntWeb holds **141,430** Malagasy ant records, **6,995** of them with images, across **60** imaged genera. Naming even the genus takes a specialist and a microscope.
- The specimens, the imaging and the determinations are curated and published by the California Academy of Sciences (San Francisco); the expertise sits outside Madagascar, and a field team waits for it.
- A photo-to-genus first pass that runs on a CPU box in Antananarivo in **~0.8 s** per image lets a local team triage, then send only the hard cases to a specialist.



One AntWeb profile shot per genus in scope (27). Sources: reports/01_explore.txt, README.md.

From GBIF to 1,236 clean images in 27 genera

141,430

AntWeb records for Madagascar on GBIF (doi:10.15468/wqmjjt); 6,995 with an image



4,354

specimens, 39 genera: workers, profile view, one image each, ≥ 10 per genus



1,297

images obtained (29.8 %): 1,160 Wikimedia Commons + 137 GBIF cache



1,236

images, **27 genera**, 8 subfamilies; 986 train / 250 held-out test

Finding 1: GBIF's image cache is empty for AntWeb

4,215 of 4,354 cache requests returned 404 (96.8 %), uniformly across years and record age. antweb.org blocks scripts (Cloudflare) and datacenter IPs (403), including GBIF's own fetcher. The working source is the 2008-2014 bulk upload to Wikimedia Commons (33,081 files), matched by specimen code.

Finding 2: caste is not in GBIF's interpreted fields

GBIF maps sex to Male / Female / Other, so every worker arrives as null: 4,540 of 5,857 specimens. The caste has to be read from the verbatim record (dwc:sex): 4,539 workers, 673 queens, 640 males, and only workers are kept.

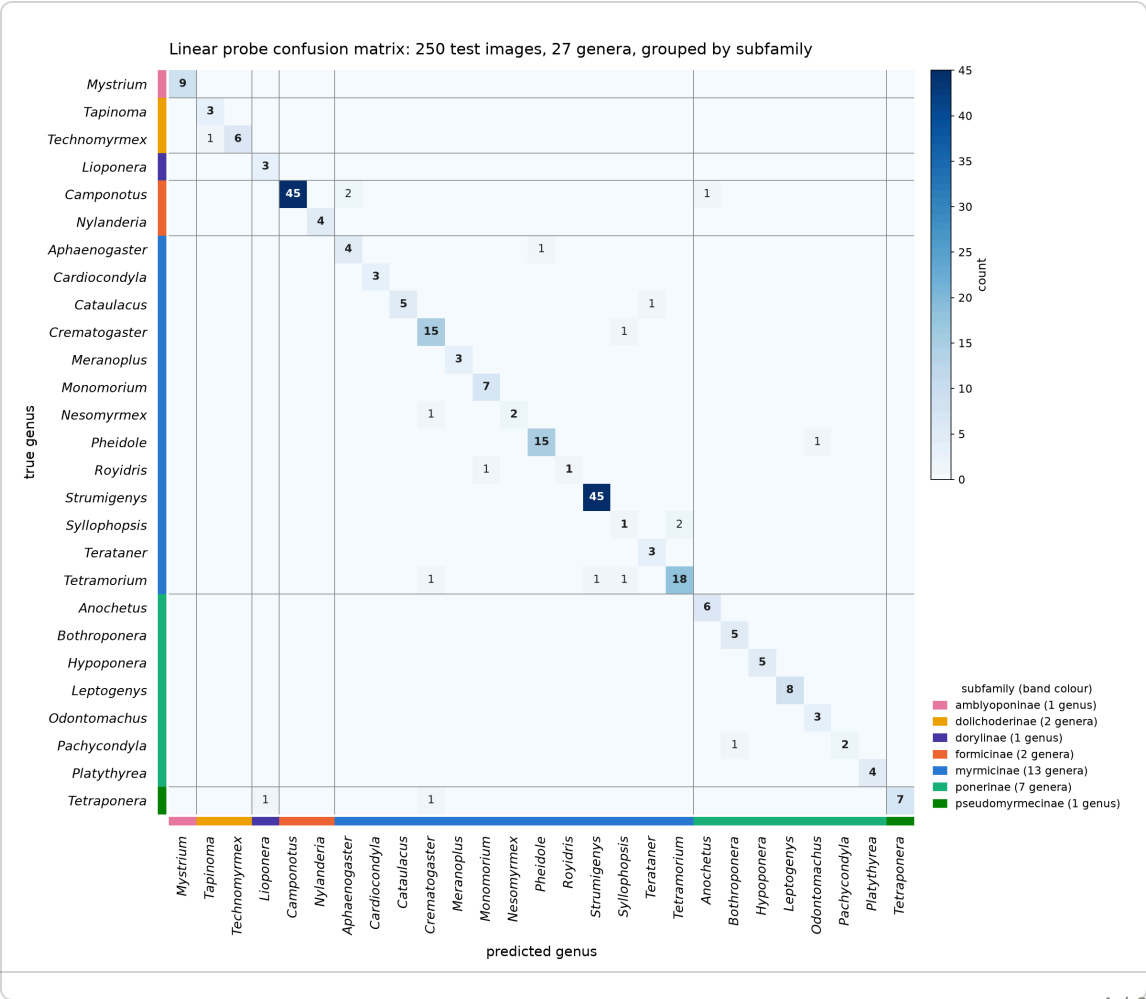
RESULTS

Linear probe on frozen BioCLIP 2 embeddings: 92.8 % top-1 on 250 held-out specimens

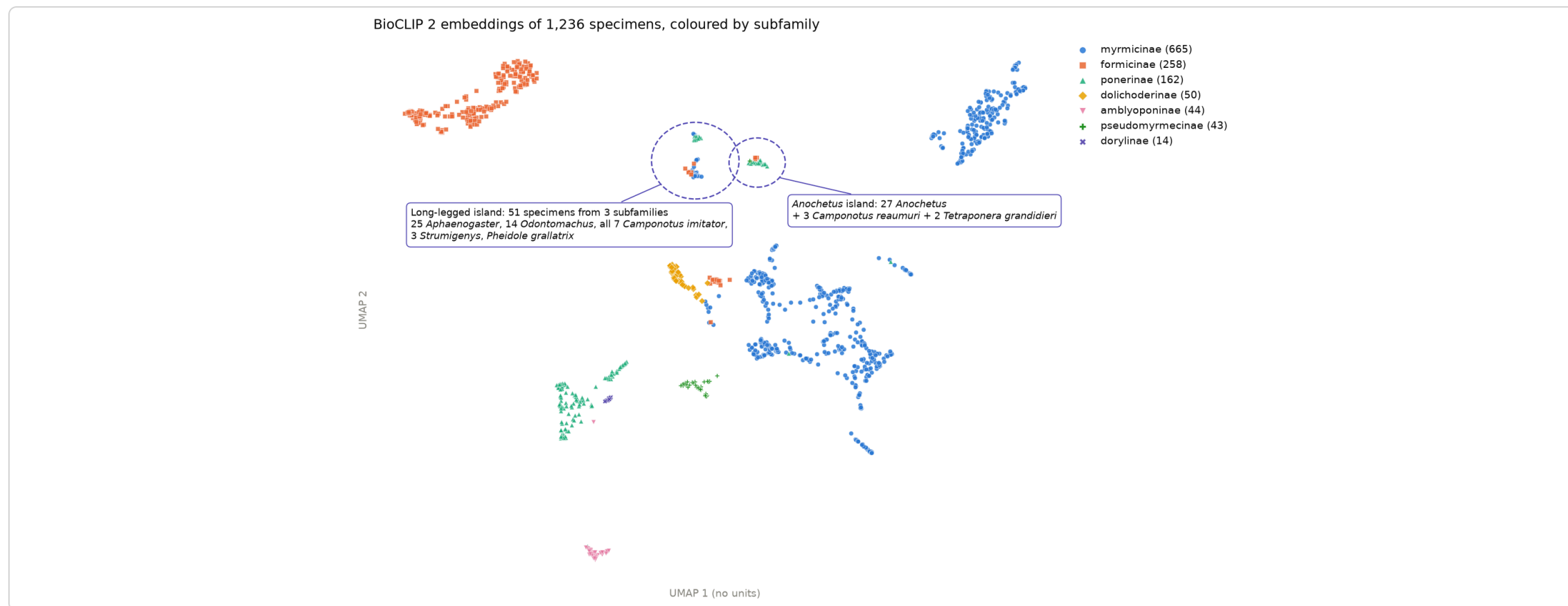
METHOD	TOP - 1	TOP - 3	MACRO - F1
Zero-shot, "a photo of {genus}, a genus of ant"	74.4 %	94.0 %	0.590
Linear probe (deployed)	92.8 %	98.4 %	0.885
Nearest neighbour, cosine	91.6 %	93.2 %	0.834

986 train / 250 test, 27 genera, split by genus (seed 42). Bare genus name as the zero-shot prompt: 32.8 % top-1. Embeddings: imageomics/bioclclip-2 ViT-L/14, 768-d, no fine-tuning.

18 errors: 12 genuine look-alikes or species unseen in training (*Sylophopsis* vs *Tetramorium*, *Royidris* vs *Monomorium*), 3 poor photos, 2 odd angles, 1 unexplained. Almost every confusion stays inside a subfamily block.



The embedding space recapitulates taxonomy, with body-plan islands



Subfamilies form clean clusters; where the model "errs", it groups *Camponotus imitator* with long-legged *Aphaenogaster* and *Odontomachus*, a resemblance a taxonomist would recognise, not noise.

Three things this codebase already supports

1

Complete the image set

70.2 % of the manifest's images are missing. `05_download_antweb.py` + `dataset_full.csv` are ready for a residential IP or an AntWeb bulk export; re-thresholding brings *Vitsika*, *Tanipone*, *Carebara* and 9 more genera into scope (39 in total).

2

Honest generalisation and an open set

Group the split by species (one flag in `03_build_dataset.py`), calibrate the probe's probabilities on the held-out set, and turn the nearest-neighbour similarity the API already returns into a "none of the 27" reject.

3

Multi-view, multi-caste embeddings

The harvest already holds 6,634 head and 5,127 dorsal media rows plus 673 queens and 640 males; `config.yaml` (`views_priority`, `castes`) and `06_embed.py` make view-wise embedding and late fusion a configuration change.